

Actividad1

August 31, 2026

```
[5]: #ACTIVIDAD 1. MARCELA GUERRERO HERNÁNDEZ

#Tratamiento de datos
import numpy as np
import pandas as pd

#Gráficos
import matplotlib.pyplot as plt

#Preprocesado y modelado
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from numpy import linalg as LA
```

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[7]: # Datos del iris.csv organizados en un arreglo numpy

df_completo = pd.DataFrame(np.array([
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[5.9, 3.0, 5.1, 1.8, "Virginica"]
]),
index = range(1, 151), # Índices del 1 al 150
columns = ['sepal.length', 'sepal.width', 'petal.length', 'petal.width', 'variety'])

# Guardamos la columna categórica en una variable aparte
variety = df_completo['variety']

# df se queda solo con las columnas numéricas
df = df_completo.drop(columns=['variety'])

```

```
df
```

```
[7]:      sepal.length sepal.width petal.length petal.width
1          5.1          3.5          1.4          0.2
2          4.9          3.0          1.4          0.2
3          4.7          3.2          1.3          0.2
4          4.6          3.1          1.5          0.2
5          5.0          3.6          1.4          0.2
..          ...          ...          ...          ...
146         6.7          3.0          5.2          2.3
147         6.3          2.5          5.0          1.9
148         6.5          3.0          5.2          2.0
149         6.2          3.4          5.4          2.3
150         5.9          3.0          5.1          1.8
```

```
[150 rows x 4 columns]
```

```
[8]: #Determinar el numero de renglones de la base de datos
index = df.index
renglones = len(index)
renglones
```

```
[8]: 150
```

```
[9]: #Estandarización de las variables
df2 = StandardScaler().fit_transform(df)
df2
```

```
[9]: array([[ -9.00681170e-01,  1.01900435e+00, -1.34022653e+00,
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```

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  1.44883158e+00],
[ 6.86617933e-02, -1.31979479e-01, 7.62758269e-01,
  7.90670654e-01]]))

```

```
[10]: #Calculo de la matriz de correlaciones para la matriz transformada
A = (1/ renglones) * np.dot(df2.T,df2)
A
```

```
[10]: array([[ 1.          , -0.11756978,  0.87175378,  0.81794113],
        [-0.11756978,  1.          , -0.4284401 , -0.36612593],
        [ 0.87175378, -0.4284401 ,  1.          ,  0.96286543],
        [ 0.81794113, -0.36612593,  0.96286543,  1.          ]])
```

```
[11]: #Entrenamiento del modelo PCA con escalado de los datos
pca_pipe = make_pipeline(StandardScaler(), PCA())
pca_pipe.fit(df)

#Se extrae el modelo entrenado del pipeline
modelo_pca = pca_pipe.named_steps['pca']
print("Eigenvalores:")
results = LA.eigvals(A)
print(results)
```

Eigenvalores:
[2.91849782 0.91403047 0.14675688 0.02071484]

```
[12]: # Porcentaje de la varianza explicada por cada nuevo componente
print("Porcentaje de varianza explicada por componente")
print(modelo_pca.explained_variance_ratio_)

#RESPUESTA DEL PASO 2 DE LA ACTIVIDAD
print("Grado de explicación de la varianza al utilizar los dos primeros_
    ↪componentes:")

#(Suma los dos primeros componentes)
print(((modelo_pca.explained_variance_ratio_[0]+modelo_pca.
    ↪explained_variance_ratio_[1])*100))
```

Porcentaje de varianza explicada por componente
[0.72962445 0.22850762 0.03668922 0.00517871]
Grado de explicación de la varianza al utilizar los dos primeros componentes:
95.81320720000164

```
[32]:
```

Eigenvectores (por renglón):

```
[32]:      sepal.length  sepal.width  petal.length  petal.width
PC1      0.521066    -0.269347     0.580413    0.564857
PC2      0.377418     0.923296     0.024492    0.066942
PC3     -0.719566     0.244382     0.142126    0.634273
PC4     -0.261286     0.123510     0.801449   -0.523597
```

```
[13]: #Proyecciones de los componentes
proyecciones = np.dot(modelo_pca.components_, df2.T)
proyecciones = pd.DataFrame(proyecciones, index = ['PC1', 'PC2', 'PC3', 'PC4'])
proyecciones = proyecciones.transpose().set_index(df.index)
proyecciones['variety'] = variety.values # <-- Volvemos a agregar la variable
↳no numérica
proyecciones
```

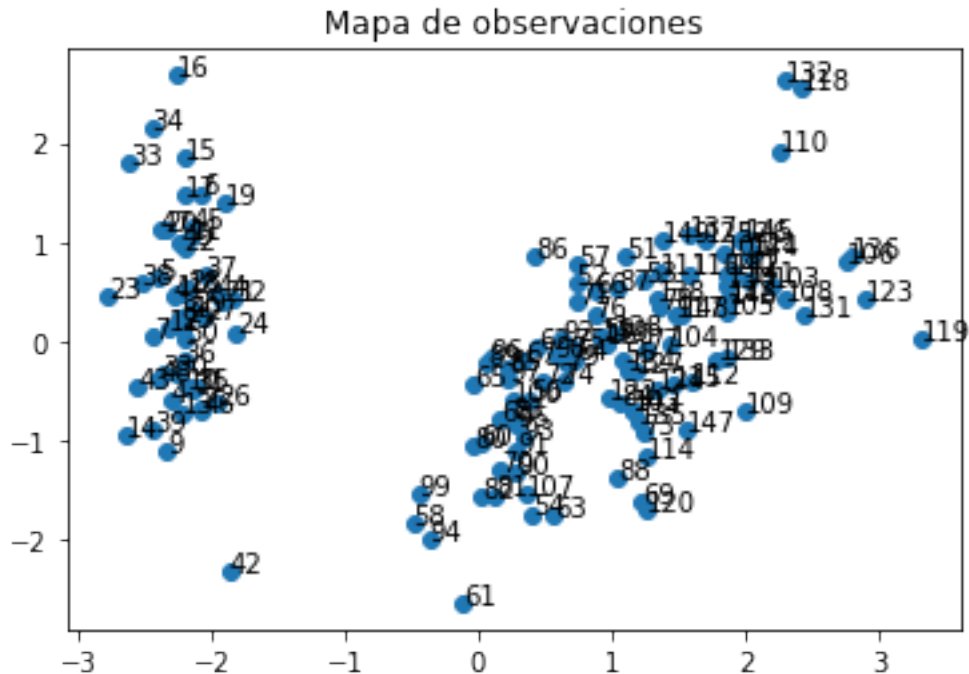
```
[13]:
```

	PC1	PC2	PC3	PC4	variety
1	-2.264703	0.480027	-0.127706	-0.024168	Setosa
2	-2.080961	-0.674134	-0.234609	-0.103007	Setosa
3	-2.364229	-0.341908	0.044201	-0.028377	Setosa
4	-2.299384	-0.597395	0.091290	0.065956	Setosa
5	-2.389842	0.646835	0.015738	0.035923	Setosa
..	...	...	...	...	...
146	1.870503	0.386966	0.256274	-0.389257	Virginica
147	1.564580	-0.896687	-0.026371	-0.220192	Virginica
148	1.521170	0.269069	0.180178	-0.119171	Virginica
149	1.372788	1.011254	0.933395	-0.026129	Virginica
150	0.960656	-0.024332	0.528249	0.163078	Virginica

[150 rows x 5 columns]

```
[14]: x = proyecciones.iloc[:,0]
y = proyecciones.iloc[:,1]
z = df.index
x = x.to_numpy()
y = y.to_numpy()
fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x,y)

for i, txt in enumerate(z):
    ax.annotate(txt, (x[i], y[i]))
```



```
[15]: x = proyecciones.iloc[:,0]
y = proyecciones.iloc[:,1]
especies = proyecciones['variety']

x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones con la variedad")

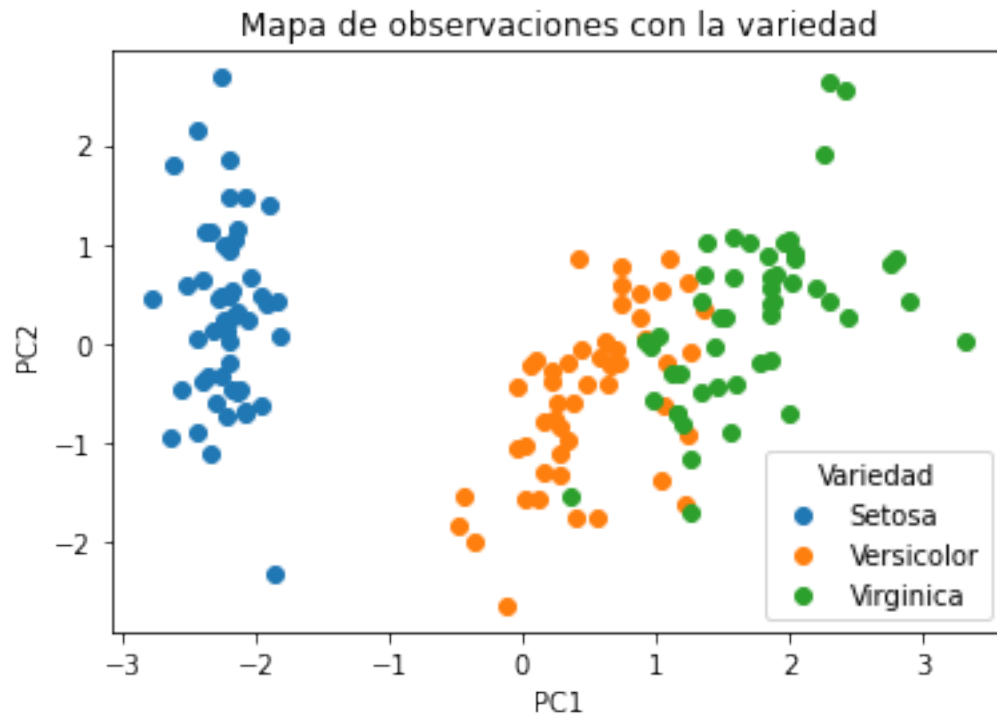
for especie in especies.unique():
    mask = (especies == especie).to_numpy()
    ax.scatter(x[mask], y[mask], label=especie)

ax.set_xlabel("PC1")
ax.set_ylabel("PC2")
ax.legend(title="Variedad")

#RESPUESTA DEL PASO 3 DE LA ACTIVIDAD
print("Se observan dos clusters principales, donde el cluster 1 se compone de
todas las setosas excepto la 42 y el segundo cluster incluye a las
variedades de versicolor, virginica y la obs 42 que pertenece al grupo de
las setosas.")
```

Se observan dos clusters principales, donde el cluster 1 se compone de todas las

setosas excepto la 42 y el segundo cluster incluye a las variedades de versicolor, virginica y la obs 42 que pertenece al grupo de las setosas.



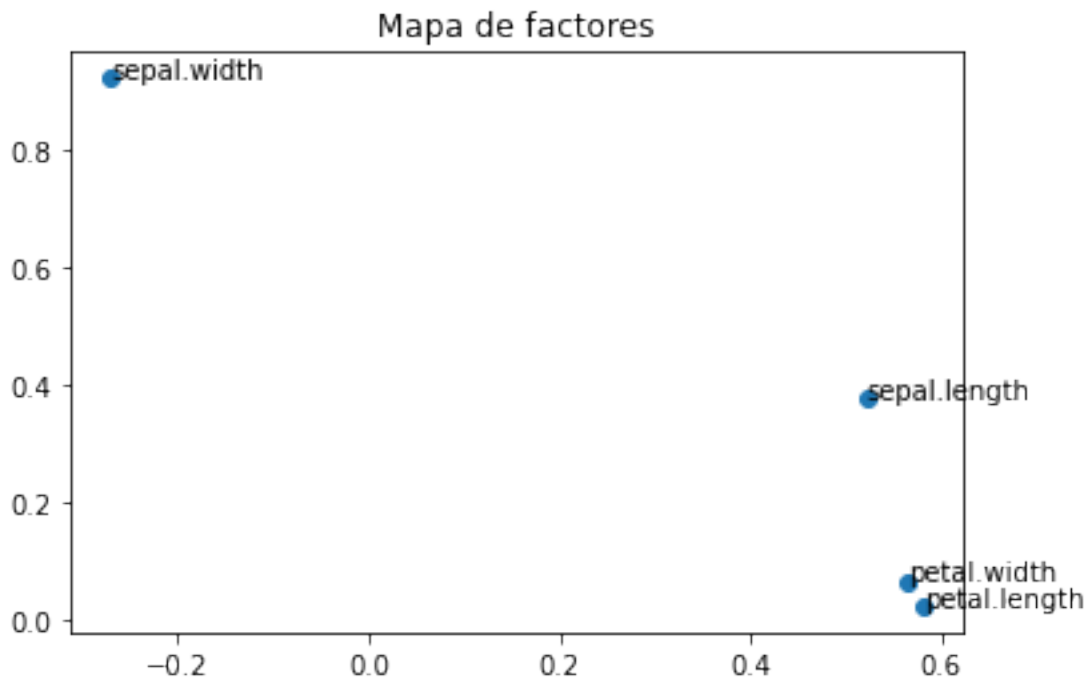
```
[16]: componentes_2 = pd.DataFrame(data= modelo_pca.components_,
                                columns = df.columns,
                                index = ['PC1', 'PC2', 'PC3', 'PC4'])
componentes_2 = componentes_2.iloc[0:2,:]
componentes_2 = componentes_2.T
componentes_2
```

```
[16]:
```

	PC1	PC2
sepal.length	0.521066	0.377418
sepal.width	-0.269347	0.923296
petal.length	0.580413	0.024492
petal.width	0.564857	0.066942

```
[18]: x = componentes_2.iloc[:,0]
y = componentes_2.iloc[:,1]
z = componentes_2.index
x = x.to_numpy()
y = y.to_numpy()
fig, ax = plt.subplots()
ax.set_title("Mapa de factores")
ax.scatter(x,y)
```

```
for i, txt in enumerate(z):
    ax.annotate(txt, (x[i], y[i]))
```



```
[19]: #Procedimiento para obtener la matriz estandarizada original
proyecciones_num = proyecciones.drop(columns=['variety']) # quitamos la
↳ columna de texto
original = np.dot(modelo_pca.components_.T, proyecciones_num.T)
original = original.T
print(original)
```

```
[[-9.00681170e-01  1.01900435e+00 -1.34022653e+00 -1.31544430e+00]
 [-1.14301691e+00 -1.31979479e-01 -1.34022653e+00 -1.31544430e+00]
 [-1.38535265e+00  3.28414053e-01 -1.39706395e+00 -1.31544430e+00]
 [-1.50652052e+00  9.82172869e-02 -1.28338910e+00 -1.31544430e+00]
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 [-5.37177559e-01  1.93979142e+00 -1.16971425e+00 -1.05217993e+00]
 [-1.50652052e+00  7.88807586e-01 -1.34022653e+00 -1.18381211e+00]
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 [-1.74885626e+00 -3.62176246e-01 -1.34022653e+00 -1.31544430e+00]
 [-1.14301691e+00  9.82172869e-02 -1.28338910e+00 -1.44707648e+00]
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```



```

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[-1.26418478e+00 -1.31979479e-01 -1.34022653e+00 -1.18381211e+00]
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[-1.02184904e+00  5.58610819e-01 -1.34022653e+00 -1.31544430e+00]
[ 1.40150837e+00  3.28414053e-01  5.35408562e-01  2.64141916e-01]
[ 6.74501145e-01  3.28414053e-01  4.21733708e-01  3.95774101e-01]
[ 1.28034050e+00  9.82172869e-02  6.49083415e-01  3.95774101e-01]
[-4.16009689e-01 -1.74335684e+00  1.37546573e-01  1.32509732e-01]
[ 7.95669016e-01 -5.92373012e-01  4.78571135e-01  3.95774101e-01]
[-1.73673948e-01 -5.92373012e-01  4.21733708e-01  1.32509732e-01]
[ 5.53333275e-01  5.58610819e-01  5.35408562e-01  5.27406285e-01]
[-1.14301691e+00 -1.51316008e+00 -2.60315415e-01 -2.62386821e-01]
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```

```

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[-2.94841818e-01 -3.62176246e-01 -8.98031345e-02 1.32509732e-01]
[ 1.03800476e+00 9.82172869e-02 3.64896281e-01 2.64141916e-01]
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[-2.94841818e-01 -1.28296331e+00 8.07091462e-02 -1.30754636e-01]
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[ 6.74501145e-01 -3.62176246e-01 3.08058854e-01 1.32509732e-01]
[ 9.16836886e-01 -1.31979479e-01 3.64896281e-01 2.64141916e-01]
[ 1.15917263e+00 -5.92373012e-01 5.92245988e-01 2.64141916e-01]
[ 1.03800476e+00 -1.31979479e-01 7.05920842e-01 6.59038469e-01]
[ 1.89829664e-01 -3.62176246e-01 4.21733708e-01 3.95774101e-01]
[-1.73673948e-01 -1.05276654e+00 -1.46640561e-01 -2.62386821e-01]
[-4.16009689e-01 -1.51316008e+00 2.38717193e-02 -1.30754636e-01]
[-4.16009689e-01 -1.51316008e+00 -3.29657076e-02 -2.62386821e-01]
[-5.25060772e-02 -8.22569778e-01 8.07091462e-02 8.77547895e-04]
[ 1.89829664e-01 -8.22569778e-01 7.62758269e-01 5.27406285e-01]
[-5.37177559e-01 -1.31979479e-01 4.21733708e-01 3.95774101e-01]
[ 1.89829664e-01 7.88807586e-01 4.21733708e-01 5.27406285e-01]
[ 1.03800476e+00 9.82172869e-02 5.35408562e-01 3.95774101e-01]
[ 5.53333275e-01 -1.74335684e+00 3.64896281e-01 1.32509732e-01]
[-2.94841818e-01 -1.31979479e-01 1.94384000e-01 1.32509732e-01]
[-4.16009689e-01 -1.28296331e+00 1.37546573e-01 1.32509732e-01]
[-4.16009689e-01 -1.05276654e+00 3.64896281e-01 8.77547895e-04]
[ 3.10997534e-01 -1.31979479e-01 4.78571135e-01 2.64141916e-01]
[-5.25060772e-02 -1.05276654e+00 1.37546573e-01 8.77547895e-04]
[-1.02184904e+00 -1.74335684e+00 -2.60315415e-01 -2.62386821e-01]
[-2.94841818e-01 -8.22569778e-01 2.51221427e-01 1.32509732e-01]
[-1.73673948e-01 -1.31979479e-01 2.51221427e-01 8.77547895e-04]
[-1.73673948e-01 -3.62176246e-01 2.51221427e-01 1.32509732e-01]
[ 4.32165405e-01 -3.62176246e-01 3.08058854e-01 1.32509732e-01]
[-9.00681170e-01 -1.28296331e+00 -4.30827696e-01 -1.30754636e-01]
[-1.73673948e-01 -5.92373012e-01 1.94384000e-01 1.32509732e-01]
[ 5.53333275e-01 5.58610819e-01 1.27429511e+00 1.71209594e+00]
[-5.25060772e-02 -8.22569778e-01 7.62758269e-01 9.22302838e-01]
[ 1.52267624e+00 -1.31979479e-01 1.21745768e+00 1.18556721e+00]
[ 5.53333275e-01 -3.62176246e-01 1.04694540e+00 7.90670654e-01]
[ 7.95669016e-01 -1.31979479e-01 1.16062026e+00 1.31719939e+00]
[ 2.12851559e+00 -1.31979479e-01 1.61531967e+00 1.18556721e+00]
[-1.14301691e+00 -1.28296331e+00 4.21733708e-01 6.59038469e-01]
[ 1.76501198e+00 -3.62176246e-01 1.44480739e+00 7.90670654e-01]
[ 1.03800476e+00 -1.28296331e+00 1.16062026e+00 7.90670654e-01]
[ 1.64384411e+00 1.24920112e+00 1.33113254e+00 1.71209594e+00]

```

```
[ 7.95669016e-01  3.28414053e-01  7.62758269e-01  1.05393502e+00]
[ 6.74501145e-01 -8.22569778e-01  8.76433123e-01  9.22302838e-01]
[ 1.15917263e+00 -1.31979479e-01  9.90107977e-01  1.18556721e+00]
[-1.73673948e-01 -1.28296331e+00  7.05920842e-01  1.05393502e+00]
[-5.25060772e-02 -5.92373012e-01  7.62758269e-01  1.58046376e+00]
[ 6.74501145e-01  3.28414053e-01  8.76433123e-01  1.44883158e+00]
[ 7.95669016e-01 -1.31979479e-01  9.90107977e-01  7.90670654e-01]
[ 2.24968346e+00  1.70959465e+00  1.67215710e+00  1.31719939e+00]
[ 2.24968346e+00 -1.05276654e+00  1.78583195e+00  1.44883158e+00]
[ 1.89829664e-01 -1.97355361e+00  7.05920842e-01  3.95774101e-01]
[ 1.28034050e+00  3.28414053e-01  1.10378283e+00  1.44883158e+00]
[-2.94841818e-01 -5.92373012e-01  6.49083415e-01  1.05393502e+00]
[ 2.24968346e+00 -5.92373012e-01  1.67215710e+00  1.05393502e+00]
[ 5.53333275e-01 -8.22569778e-01  6.49083415e-01  7.90670654e-01]
[ 1.03800476e+00  5.58610819e-01  1.10378283e+00  1.18556721e+00]
[ 1.64384411e+00  3.28414053e-01  1.27429511e+00  7.90670654e-01]
[ 4.32165405e-01 -5.92373012e-01  5.92245988e-01  7.90670654e-01]
[ 3.10997534e-01 -1.31979479e-01  6.49083415e-01  7.90670654e-01]
[ 6.74501145e-01 -5.92373012e-01  1.04694540e+00  1.18556721e+00]
[ 1.64384411e+00 -1.31979479e-01  1.16062026e+00  5.27406285e-01]
[ 1.88617985e+00 -5.92373012e-01  1.33113254e+00  9.22302838e-01]
[ 2.49201920e+00  1.70959465e+00  1.50164482e+00  1.05393502e+00]
[ 6.74501145e-01 -5.92373012e-01  1.04694540e+00  1.31719939e+00]
[ 5.53333275e-01 -5.92373012e-01  7.62758269e-01  3.95774101e-01]
[ 3.10997534e-01 -1.05276654e+00  1.04694540e+00  2.64141916e-01]
[ 2.24968346e+00 -1.31979479e-01  1.33113254e+00  1.44883158e+00]
[ 5.53333275e-01  7.88807586e-01  1.04694540e+00  1.58046376e+00]
[ 6.74501145e-01  9.82172869e-02  9.90107977e-01  7.90670654e-01]
[ 1.89829664e-01 -1.31979479e-01  5.92245988e-01  7.90670654e-01]
[ 1.28034050e+00  9.82172869e-02  9.33270550e-01  1.18556721e+00]
[ 1.03800476e+00  9.82172869e-02  1.04694540e+00  1.58046376e+00]
[ 1.28034050e+00  9.82172869e-02  7.62758269e-01  1.44883158e+00]
[-5.25060772e-02 -8.22569778e-01  7.62758269e-01  9.22302838e-01]
[ 1.15917263e+00  3.28414053e-01  1.21745768e+00  1.44883158e+00]
[ 1.03800476e+00  5.58610819e-01  1.10378283e+00  1.71209594e+00]
[ 1.03800476e+00 -1.31979479e-01  8.19595696e-01  1.44883158e+00]
[ 5.53333275e-01 -1.28296331e+00  7.05920842e-01  9.22302838e-01]
[ 7.95669016e-01 -1.31979479e-01  8.19595696e-01  1.05393502e+00]
[ 4.32165405e-01  7.88807586e-01  9.33270550e-01  1.44883158e+00]
[ 6.86617933e-02 -1.31979479e-01  7.62758269e-01  7.90670654e-01]]
```

[31]: **#RESPUESTA DEL PASO 3 DE LA ACTIVIDAD: CASOS ESPECIFICOS**

**#Las obs elegidas fueron tomadas del mapa de observaciones de cada cluster y se
 espera que sean distintas entre sí.**

#Se espera que sea Setosa

```

Obs_Cluster1 = df_completo.iloc[14]

#Se espera que sea Versicolor o Virginica
Obs_Cluster2 = df_completo.iloc[118]

print(Obs_Cluster1)
print(Obs_Cluster2)

#RESPUESTA
print("Las dos flores son muy diferentes entre sí. La primera es una Setosa, que tiene pétalos muy pequeños y sépalos anchos, lo que la hace ver compacta y pequeña.")
print("La segunda es una Virginica, que tiene pétalos mucho más grandes y sépalos más largos pero delgados, dándole una apariencia más alargada y grande.")
print("La diferencia más notable está en el tamaño de los pétalos: los de la Virginica son mucho más grandes que los de la Setosa. Estas claras diferencias en todas sus medidas explican por qué pertenecen a grupos distintos en el análisis.")

```

```

sepal.length    5.8
sepal.width     4.0
petal.length    1.2
petal.width     0.2
variety         Setosa
Name: 15, dtype: object
sepal.length    7.7
sepal.width     2.6
petal.length    6.9
petal.width     2.3
variety         Virginica
Name: 119, dtype: object

```

Las dos flores son muy diferentes entre sí. La primera es una Setosa, que tiene pétalos muy pequeños y sépalos anchos, lo que la hace ver compacta y pequeña. La segunda es una Virginica, que tiene pétalos mucho más grandes y sépalos más largos pero delgados, dándole una apariencia más alargada y grande.

La diferencia más notable está en el tamaño de los pétalos: los de la Virginica son mucho más grandes que los de la Setosa. Estas claras diferencias en todas sus medidas explican por qué pertenecen a grupos distintos en el análisis.

[]:

[]: